

LearnLoop – Functional & Non-Functional Requirements

Prepared based on the Initial Requirement Document (IRD).

Functional Requirements

ID	Requirement
FR1	Users shall register and log in securely using role-based authentication.
FR2	The system shall support three user roles: Student, Teacher, and Administrator.
FR3	Students shall access learning modules and educational resources.
FR4	Teachers shall create, update, and manage learning modules and study materials.
FR5	Students shall upload and share study resources such as notes, PDFs, presentations, and useful links.
FR6	Students shall ask questions and answer queries through a collaborative discussion forum.
FR7	The system shall implement a credit-based reward mechanism for learning activities and knowledge contributions.
FR8	Students shall earn credits by completing learning modules, quizzes, sharing resources, and helping other learners.
FR9	Students shall redeem earned credits for premium resources, AI features, certificates, or special activities.
FR10	The system shall provide an AI-powered chatbot to answer academic queries and assist learners.
FR11	The system shall recommend personalized learning paths based on user interests, learning history, and performance.
FR12	Teachers shall create and manage quizzes, and students shall receive instant evaluation and feedback.
FR13	The system shall track learner progress, achievements, learning history, and credit transactions.
FR14	The system shall provide gamification features including badges, achievements, learning streaks, and leaderboards.
FR15	Administrators shall manage users, teachers, learning resources, credits, reports, and system settings through an administrative dashboard.

Non-Functional Requirements

ID	Category	Requirement
NFR1	Performance	The system should respond within 10 seconds for common operations under normal load.
NFR2	Security	The system shall implement password encryption, JWT authentication, role-based access control, and secure data storage.
NFR3	Reliability	The system shall ensure data consistency with backup and recovery mechanisms.
NFR4	Availability	The platform should be available 99% of the time, excluding scheduled maintenance.
NFR5	Usability	The application shall provide an intuitive, responsive, and user-friendly interface for Students, Teachers, and Administrators.
NFR6	Scalability	The system shall support increasing numbers of users, learning resources, and AI interactions without significant performance degradation.
NFR7	Maintainability	The application shall use a modular architecture to simplify maintenance and future enhancements.
NFR8	Portability	The platform shall be accessible through modern web browsers and responsive on desktops, tablets, and mobile devices.
NFR9	Auditability	The system shall maintain logs of user activities, credit transactions, and administrative actions.
NFR10	Data Integrity	The system shall prevent duplicate records and ensure the accuracy and consistency of stored data.